

VITECH ENVIRO SYSTEMS AI ENGINEERING DATABASE

PAINT BOOTH – SOURCE-BASED DATABASE v02

This revision converts the previous framework into a source-based VITECH paint-booth dataset using the newly supplied VITECH blower table, paint-booth BOMs and a dry-type cross-draft GA drawing.

Data classification: Values directly present in supplied VITECH documents are treated as source data. Derived relationships are labelled as observed/derived. Missing engineering basis is not silently filled with generic assumptions.

1. VITECH PAINT BOOTH TYPES IDENTIFIED

Family	Configuration	Source-derived information
Wet Paint Booth	Front Open	VITECH model series VT/1.5 to VT/4.5/DTPB/OP; W=1500–4500; D=1500; H=2425; face velocity 0.5; capture velocity 1.0; static shown as 5.
Wet Paint Booth	Enclosed	VT/1.5 to VT/7.5/DTPB/CL; D=2250; H=2425; face velocity 0.5; capture velocity 1.0; static shown as 5.
Dry Paint Booth	Front Open – 2-row filter	VT/1.5 to VT/4.5/DTPB/OP; D=1500; H=2425; velocity 0.5; static shown as 3.
Dry Paint Booth	Enclosed – 2-row filter	VT/1.5 to VT/4.5/DTPB/CL; D=2250; H=2425; velocity 0.5; static shown as 4.
Dry Paint Booth	Front Open – 3-row filter	VT/1.5 to VT/4.5/DTPB/OP; D=1500; H=2425; velocity 0.5 in source table; airflow values imply ~0.70 m/s over 1.5 m
Dry Paint Booth	Enclosed – 3-row filter	VT/1.5 to VT/4.5/DTPB/CL; D=2250; H=2425; velocity 0.5 in source table; airflow values imply ~0.70 m/s over 1.5 m

2. WET PAINT BOOTH – SOURCE TABLE

Model	W	D	H	Static	Face V	Capture V	CMH	CFM	HP
VT/1.5/DTPB/OP	1500	1500	2425	5	0.5	1.0	3645	2144	5
VT/2.25/DTPB/OP	2250	1500	2425	5	0.5	1.0	5468	3216	5
VT/3.0/DTPB/OP	3000	1500	2425	5	0.5	1.0	7290	4288	7.5
VT/3.75/DTPB/OP	3750	1500	2425	5	0.5	1.0	9113	5360	7.5
VT/4.5/DTPB/OP	4500	1500	2425	5	0.5	1.0	10935	6432	10
VT/1.5/DTPB/CL	1500	2250	2425	5	0.5	1.0	3645	2144	5
VT/2.25/DTPB/CL	2250	2250	2425	5	0.5	1.0	5468	3216	5
VT/3.0/DTPB/CL	3000	2250	2425	5	0.5	1.0	7290	4288	7.5
VT/3.75/DTPB/CL	3750	2250	2425	5	0.5	1.0	9113	5360	7.5
VT/4.5/DTPB/CL	4500	2250	2425	5	0.5	1.0	10935	6432	10
VT/7.5/DTPB/CL	7500	2250	2425	5	0.5	1.0	18225	10721	—

3. DRY TYPE PAINT BOOTH – 2 ROW FILTER

Model	W	D	H	Static	V	CMH	CFM	HP	Config
VT/1.5/DTPB/OP	1500	1500	2425	3	0.5	4050	2382	2	Front Open
VT/2.25/DTPB/OP	2250	1500	2425	3	0.5	6075	3574	3	Front Open
VT/3.0/DTPB/OP	3000	1500	2425	3	0.5	8100	4765	5	Front Open
VT/3.75/DTPB/OP	3750	1500	2425	3	0.5	10125	5956	5	Front Open
VT/4.5/DTPB/OP	4500	1500	2425	3	0.5	12150	7147	7.5	Front Open
VT/1.5/DTPB/CL	1500	2250	2425	4	0.5	4050	2382	2	Enclosed
VT/2.25/DTPB/CL	2250	2250	2425	4	0.5	6075	3574	3	Enclosed
VT/3.0/DTPB/CL	3000	2250	2425	4	0.5	8100	4765	5	Enclosed
VT/3.75/DTPB/CL	3750	2250	2425	4	0.5	10125	5956	5	Enclosed

Model	W	D	H	Static	V	CMH	CFM	HP	Config
VT/4.5/DTPB/CL	4500	2250	2425	4	0.5	12150	7147	7.5	Enclosed

4. DRY TYPE PAINT BOOTH – 3 ROW FILTER

Model	W	D	H	Static	V shown	CMH	CFM	HP	Config
VT/1.5/DTPB/OP	1500	1500	2425	3	0.5	5670	3335	5	Front Open
VT/2.25/DTPB/OP	2250	1500	2425	3	0.5	8505	5003	7.5	Front Open
VT/3.0/DTPB/OP	3000	1500	2425	3	0.5	11340	6671	7.5	Front Open
VT/3.75/DTPB/OP	3750	1500	2425	3	0.5	14175	8338	10	Front Open
VT/4.5/DTPB/OP	4500	1500	2425	3	0.5	17010	10006	10	Front Open
VT/1.5/DTPB/CL	1500	2250	2425	4	0.5	5670	3335	5	Enclosed
VT/2.25/DTPB/CL	2250	2250	2425	4	0.5	8505	5003	7.5	Enclosed
VT/3.0/DTPB/CL	3000	2250	2425	4	0.5	11340	6671	7.5	Enclosed
VT/3.75/DTPB/CL	3750	2250	2425	4	0.5	14175	8338	10	Enclosed
VT/4.5/DTPB/CL	4500	2250	2425	4	0.5	17010	10006	10	Enclosed

5. OBSERVED AIRFLOW FORMULAS

For the 2-row dry table, the listed CMH matches the 0.5 m/s calculation using a 1500 mm high effective opening/filter height:

$$\text{CMH} = W(\text{m}) \times 1.5(\text{m}) \times 0.5(\text{m/s}) \times 3600$$

$$\text{CFM} = \text{CMH} \times 0.588578$$

Example: $W=3.0 \text{ m} \rightarrow 3.0 \times 1.5 \times 0.5 \times 3600 = 8100 \text{ CMH} \rightarrow \text{approximately } 4765 \text{ CFM}$.

For the 3-row table, the listed airflow is 1.4× the 2-row airflow at the same width. If the effective height is 1.5 m, that corresponds to approximately 0.70 m/s. The source table itself displays velocity as 0.5, so the engineering basis of the 1.4 factor must be confirmed.

For the wet booth table, the listed airflow is 0.90× the simple 0.5 m/s × 1.5 m opening calculation. Record this as an **observed 0.90 factor**, not yet as a confirmed VITECH design constant.

6. VITECH BLOWER SPECIFICATION MATRIX

Static 2*

HP	CFM
2	2500
3	4000
5	8500
7.5	14000
10	18000
12.5	20000
15	25000
20	37000

Static 3*

HP	CFM
2	2300
3	3200
5	6000
7.5	10000

HP	CFM
10	12000
12.5	13500
15	18000
20	25000

Static 4*

HP	CFM
2	1600
3	2200
5	3700
7.5	6500
10	9000
12.5	11500
15	14500
20	19500

Static 5*

HP	CFM
2	1200
3	1900
5	3600
7.5	6000
10	7200
12.5	9000
15	12000
20	16000

Static 6*

HP	CFM
2	1000
3	1600
5	2900
7.5	4500
10	6000
12.5	7500
15	10000
20	13000

Static 8*

HP	CFM
2	700
3	1100
5	2200

HP	CFM
7.5	3300
10	4400
12.5	5500
15	6600
20	9000

Important: The supplied blower matrix labels static pressure as 2, 3, 4, 5, 6 and 8, but the unit is not explicitly stated in the source image. Do not train the AI to assume the unit until VITECH confirms it.

7. HISTORICAL DRY BOOTH BOM – GAG INTERNATIONAL

Project: P1010-2025-26-GAG-DTPB-OP; Date: 07.01.2026; Dry Type Paint Booth – Front Open.

Item	VITECH source BOM specification	Qty	Use/remark
CR Sheet	18 SWG, 2500×900×1.2 mm	22 Nos	Booth panels
CR Sheet	16 SWG, 2500×1250×1.6 mm	2 Nos	—
MS Plate	750×750×6 mm	1 No	—
MS Channel ISMC	75×40×6000(L)×6T	5 Nos	Structure
MS Square Tube SHS	40×40×6000(L)×2T	4 Nos	Structure
MS Equal Angle	35×35×6000(L)×6T	1 No	Light fixing
MS Flat ISF	40×6000(L)×6T	1 No	—
M10 MS Rod U-clamp	150H×50U	5 Nos	Job hanging
Weld mesh + primer	25×25×3 Dia, 3000×1400	54.2304 Sq.ft	Filter support
Blower	CLP-3-5-6000 Direct Drive; MS casting powder coated; AI impeller	1 No	Blower
Motor	5 HP / FLP / IE3 / 1440 / flange / VFD start	1 No	Motor
LED Light	FLP	2 Nos	Lighting
Control Panel	1×5 HP blower + 2×LED + VFD	1 No	Controls
Neoprene gasket	5 mm thick	1 No	As per drawing
Paper filter	1500×0.7(H) m	4 Nos	Standard
Fasteners	GI M8×20 / M8×25 / M8×60 / M10×40 / M6×15	See BOM	Assembly
Sealant	GP-101	2 Nos	Gap arrest
Painting	Primer + Enamel	1200 Sq.ft	With accessories

8. HISTORICAL GA CROSS-CHECK

The supplied dry-type cross-draft drawing states: non-pressurized cross-draft dry-type paint booth; inner size 3000(W) × 2000(D) × 2425(H) mm; exhaust blower with FLP motor, 1 set, 5.0 HP; exhaust duct outside, straight. The drawing also shows paint arrestor filter and blower with motor.

This drawing provides a valuable project-level reference for linking a 3000 × 2000 × 2425 mm booth to a 5 HP blower. It should be stored as a **historical project configuration**, not automatically generalized as the standard for every 3000 mm booth.

9. WATER CURTAIN PAINT BOOTH – HISTORICAL BOM

Project: P1015-25-2026, M/s Yonex, dated 08.04.2026; Water Curtain Paint Booth – 0.9 m W.

Category	Source data
Main MOC	SS304 10 SWG 2B, 3000×1500 – 2 Nos; SS304 18 SWG 2B, 2500×1250 – 8 Nos
Structure	SS304 square tube 40×40×1.5×6000 – 1 No; MS C-channel 75×40×6×6000 – 2 Nos; MS L-angle 35×35×5×6000 – 2 Nos
Blower	CLP-5-2-1200; Static 5; Flow 1200 CFM; direct drive; square mouth; MS powder coated; AI/other details as stated in BOM
Motor	CG, 2 HP, 1440 RPM, flange, IE2, NFLP, VFD

Category	Source data
Sludge pump	Kirloskar, 1 HP, NFLP, CI
Lighting	40 W FLP LED – 1 No
Perforated sheet	SS304, 3 mm, Ø3 mm / 6 mm pitch; 400×400 and 400×600
Pipeline	SS304/PVC arrangement including 1 in, 1.5 in and associated fittings
Valves	PP/PVC ball valves; sizes as specified in BOM
Control panel	2 HP blower with VFD + 1 HP pump + 40 W LED with controls

10. PIPELINE / COMPONENT DATABASE – OBSERVED

The newly supplied pipeline BOM contains SS304 flanges, reducers, threaded couplings, valves and PVC Schedule 40 piping/fittings, including 4 in, 2 in, 1 in and 1/2 in lines. It also includes U-clamps, PVC paste, Chellack, air hose and threaded couplings.

This should be linked to the **Water Curtain / Wet Paint Booth** subassembly as a component library, while project-specific quantities remain project BOM data.

11. AI SELECTION LOGIC – SOURCE-BASED

Customer requirement → Identify wet/dry → Front-open/enclosed → Required W/D/H → Select filter configuration → Calculate/lookup airflow → Determine static-pressure requirement → Select blower from VITECH matrix → Check motor → Retrieve BOM template → Adjust dimensions/quantities → Generate GA/specification → Engineering approval

Example source-based lookup: a 3000 mm dry front-open booth with the 2-row filter table gives 8100 CMH / 4765 CFM and 5 HP at static 3 in the supplied table. A historical 3000×2000×2425 cross-draft GA/BOM also uses a 5 HP blower. These are compatible records but should remain separately traceable to their source documents.

12. DATA QUALITY / OPEN ITEMS

Item	Status	AI treatment
Wet booth face velocity 0.5 m/s	Source table	Confirmed source value for listed models
Wet booth capture velocity 1.0 m/s	Source table	Confirmed source value for listed models
Wet booth 0.90 airflow factor	Derived from table	Observed; engineering basis to confirm
Dry 2-row velocity 0.5 m/s	Source table + matches formula	Strong source rule
Dry 3-row airflow 1.4× 2-row	Derived from table	Observed; basis to confirm
Static pressure units	Not stated in supplied blower image	Do not assume
Blower selection curve/model relation	Partial source matrix + project BOM	Need approved fan curves/model master
Filter dimensions/media/pressure drop	Partial BOM data	Need complete filter master
Standard structure rules	Historical BOMs	Need more projects before declaring standard
Cost/margin/markup	Not present in supplied files	Still required

13. REQUIRED MASTER RECORD FORMAT FOR SPARK INVENTIVE

Product ID | Type | Model | Configuration | W | D | H | Filter Rows | Face Velocity | Capture Velocity | Static Pressure | Static Unit | CMH | CFM | Blower Model | Motor HP | Motor RPM | Filter Type | Filter Size | Filter Qty | MOC | BOM ID | GA ID | Calculation ID | Source Document | Revision | Confidence

Every value should carry a source/revision link. Derived values should store the formula and source inputs. Historical project values should not automatically become universal standards.

14. NEXT DATA TO ADD

The strongest next inputs are: approved blower/fan curves and model list; paint-arrestor/paper-filter datasheets; more dry/wet booth BOMs; booth calculation sheets; water-curtain calculations; duct calculations; and costing sheets. These will allow the AI database to move from a lookup catalogue to a true VITECH engineering selection engine.

15. SOURCE DOCUMENT REGISTER

Source	Content used
6019125f-b9b9-4fa7-97d0-6dcea1bc6274.xlsx	Dry Type Paint Booth – Front Open BOM, GAG International, project P1010-2025-26-GAG-DTPB-OP
0ebb5f09-04cf-41ef-84c9-bf0580112606.xlsx	Water Curtain Paint Booth BOM, Yonex, project P1015-25-2026
be12b76e-4501-4a21-99d9-01220a2d1dee.xlsx	Pipeline BOM / pending pipeline components
9673a185-832c-498b-988b-158b059b675f.pdf	Dry Type Paint Booth – Cross Draft GA; 3000×2000×2425 inner size; 5 HP blower